

Capital Structure Planning

Capital Structure

- When a business finances a part of its operations by debt, we say that the business employs financial leverage or that it is a *levered firm*. If a business does not employ debt, we say that it is an all-equity firm (unlevered firm) and has no leverage.

Firm's Capital Structure Ratios

- The **debt/equity ratio** can be larger than 1 for a firm that employs a high level of debt.

$$\text{Debt / equity Ratio} = \frac{\text{Book value of long term debt}}{\text{Book value of equity}}$$

- The **debt/value** of the firm ratio mostly might not be larger than 1.

$$\text{Debt / value of firm Ratio} = \frac{\text{Market value of long term debt}}{\text{Market value of equity}}$$

Value of the Firm

- Assume a firm issues long-term debt, denoted by D , and equity, denoted by E . The firm's total market value, denoted by V , is the value of all its assets, which belong to bondholders and stockholders together. V is the sum of the market of component D and E .
- If a firm issues no debt, it is an unlevered firm. In an all-equity firm $D = 0$ and $V = E$.
- Net Operating Income (NOI) is the firm's earnings before it distributes any of its income as interest as dividends or as taxes. Because it is measured before any interest is deducted, NOI is determined by the firm's investments and not by its financial leverage.
- NOI, hence adds value to the firm. The NOI is also called EBIT (Earnings Before Interest and Taxes).

Capital Structure Risk

- A firm using debt in its capital has to pay fixed interest charges and the lack of ability to pay fixed interest increases the risk of liquidation.
 - Risk due to debt commitment
 - Risk due to variability in earnings
- Financial Risk (Financial leverage)
- Operating Risk (Operating leverage)
- Combined Risk (Total Leverage)

Operating and Financial Leverage

Income Statement		
	(INR Million)	
Sales	160000	
Fixed cost	60000	
Variable cost	64000	
Operating return	36000	EBIT
Interest	12000	
Profit before tax	24000	PBT
Taxes	7200	
Earnings	16800	
Number of shares	10000	
Earnings per share	1.68	EPS



Operating
Leverage



Financial
Leverage

Optimal Leverage

- Firm's goal is to maximize its value.
 - A given invested capital and a given number of shares outstanding, this goal is equivalent to maximizing the firm's stock price. Suppose a firm raises INR 1000 by issuing 100 Shares at INR 10 per share. In this case the firm's market value will be INR 1000.
 - Alternatively, if the firm finances its operations by issuing INR 500 of debt and 50 shares of stock. If the market prefers this financial mix to all-equity, the firm may be able to sell the stock for a higher price, say, INR 11 per share. In this case the firm's value will be INR 1050. The firm's value increase and when the value of the firm increases, so is the expectation from the stock price.
- Level of financial leverage that maximizes the firm's market value is the optimal financial leverage or the optimal capital structure.

Financial Leverage

- Financial leverage arises when a firm finances its assets with debt.
- Debt becomes a liability for the firm, and it must pay interest on that debt.
- A company will only take on significant amounts of debt when it believes that return on assets (ROA) will be higher than the interest on the loan.

Financial Leverage Measures

- Measurement of financial leverage can be based on the capacity of the firm to finance its interest obligations out of its operating profits. In other words the degree of financial leverage can also be computed as percentage change in taxable income over percentage change in operating income.
- Two measures of financial leverage are:

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{PBT}}$$

EBIT = Earnings before Interest and Tax (Operating profit)

PBT = Profit before Tax

$$\text{Degree of Financial Leverage} = \frac{\Delta \text{EPS}}{\Delta \text{EBIT}}$$

EPS = Earnings per Share

Interpreting Financial Leverage

- Evaluates the impact of debt equity structure of the firm
- Determines the allocation of benefits to the shareholders
- High (low) financial leverage implies high (low) debt commitment

Interpreting Degree of Financial Leverage

- Percentage change in the EPS for a 1% change in the EBIT.
- Debt increases the variability of EPS as this increases the degree of financial leverage.

Example

	1st Year (million)	2nd Year (million)
Equity capital (n shares millions @ INR 10)	2000	2000
10% Debentures	2000	4000
Earnings before Interest & Tax	400	900
Tax (@ 30%)	60	150
Financial Leverage (EBIT/PBT) (Ratio)	$(400/200)=2$	$(900/500)=1.8$
Degree of Financial Leverage (% change in EPS/ % change in EBIT)		$(150\%/125\%)=1.2$

Implications of Financial Leverage

- Measure of financial risk has reduced over time. A higher quantum of interest burden will decrease the denominator and hence increase the financial leverage measure. Thus, an increase in financial leverage or higher financial leverage indicates higher risk for the business.
- On the other hand in this example, the financial leverage has declined from 2 to 1.8 thus indicating relatively lesser interest burden on the firm. Hence this shows a favorable financial risk in terms of the second year of operation.
- Degree of financial leverage is 1.2

Tata Consultancy Services

TCS	2020	2021	2022	2023	2024
Operating Income	3,85,810	4,24,820	4,84,440	5,42,370	5,93,110
Operating Income Growth	3%	10.11%	14.03%	11.96%	9.36%
EBT, Incl. Unusual Items	4,22,480	4,37,600	5,16,870	5,69,070	6,19,970
EBT, Incl. Unusual Items Growth	1.65%	3.58%	18.11%	10.10%	8.94%
Basic EPS - Continuing Operations	86.19	86.71	103.62	115.19	125.88
Basic EPS - Continuing Operations Growth	3.78%	0.61%	19.50%	11.16%	9.29%
Financial Leverage	0.91	0.97	0.94	0.95	0.96
Degree of Financial Leverage	1.26	0.06	1.39	0.93	0.99

Operating Leverage

- Operating leverage is the application of fixed costs and variable costs for contributing towards return on assets.
- Ratio of fixed and variable costs that a firm uses determines the amount of operating leverage employed.
 - A company with a greater ratio of fixed to variable costs is said to be using more operating leverage.
 - If a firm's variable costs are higher than its fixed costs, it would be using less operating leverage.
- Business expenses ratio contributing to sales is a factor in how much leverage it employs.
 - A firm with few sales and high margins is said to be highly leveraged.
 - A firm with a high volume of sales and lower margins is said to be less leveraged.
- While financial leverage necessarily arises due to the presence of debt in capital structure, a firm has an inherent risk in terms of operating activities.
- Operating leverage can be defined as the relationship between contribution and operating profit.

Operating Leverage Measures

- Contribution is the excess of sales income over all direct expenses (variable costs).
- Degree of operating leverage can also be computed as the relationship between percentage change in operating income to percentage change in sales income.

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}}$$

Contribution = Sales - Direct expenses

EBIT = Earnings before Interest and Tax (Operating Profit)

$$\text{Degree of Operating Leverage} = \frac{\Delta \text{EBIT}}{\Delta \text{Sales}}$$

Interpretation of Operating Leverage

- Evaluates the impact on the operating structure of the firm
- Determines return from operations
- High (low) operating leverage implies high (low) fixed cost

Interpretation of Degree of Operating Leverage

- A one percentage change in sales is expected to bring a $x\%$ change in operating profit.
- Higher degree of operating leverage associates more sensitivity of earnings to operating profit.
- Closer the firm operates to its break even point, the higher will the absolute value of degree of operating leverage.
- Degree of operating leverage measures business risk through the sensitivity of operating profits to sales..

Example

	1st Year (million)	2nd Year (million)
Sales	800	1650
Variable cost	300	300
Fixed cost	100	450
Contribution (sales - Variable cost)	500	1350
EBIT (Contribution - Fixed cost)	400	900
Operating Leverage (Contribution/EBIT) (Ratio)	1.25	1.50
Degree of Operating Leverage (% change in EBIT/ % change in Sales)		(125%/106.25%)=1.18

Implications of Operating Leverage

- Operating leverage or operating risk has increased over the year.
- A high operating leverage indicates a higher business risk for the firm.
- Operating risk measure is 1.25 in the 1st year of operation, but it has increased to 1.50 in the 2nd year indicating that the operating risk has increased with increase in volume.
- Degree of operating leverage is 1.18.

Tata Consultancy Services

TCS	2020	2021	2022	2023	2024
Total Revenues	15,69,490	16,41,770	19,17,540	22,54,580	24,08,930
Total Revenues Growth	7.16%	4.61%	16.80%	17.58%	6.85%
Cost of Revenue	8,78,570	9,32,760	10,87,170	12,94,030	14,38,330
Gross Profit	6,90,920	7,09,010	8,30,370	9,60,550	9,70,600
Operating Income	3,85,810	4,24,820	4,84,440	5,42,370	5,93,110
Operating Income Growth	3%	10.11%	14.03%	11.96%	9.36%
Operating Leverage	1.79	1.67	1.71	1.77	1.64
Degree of Operating Leverage	0.42	2.19	0.84	0.68	1.37

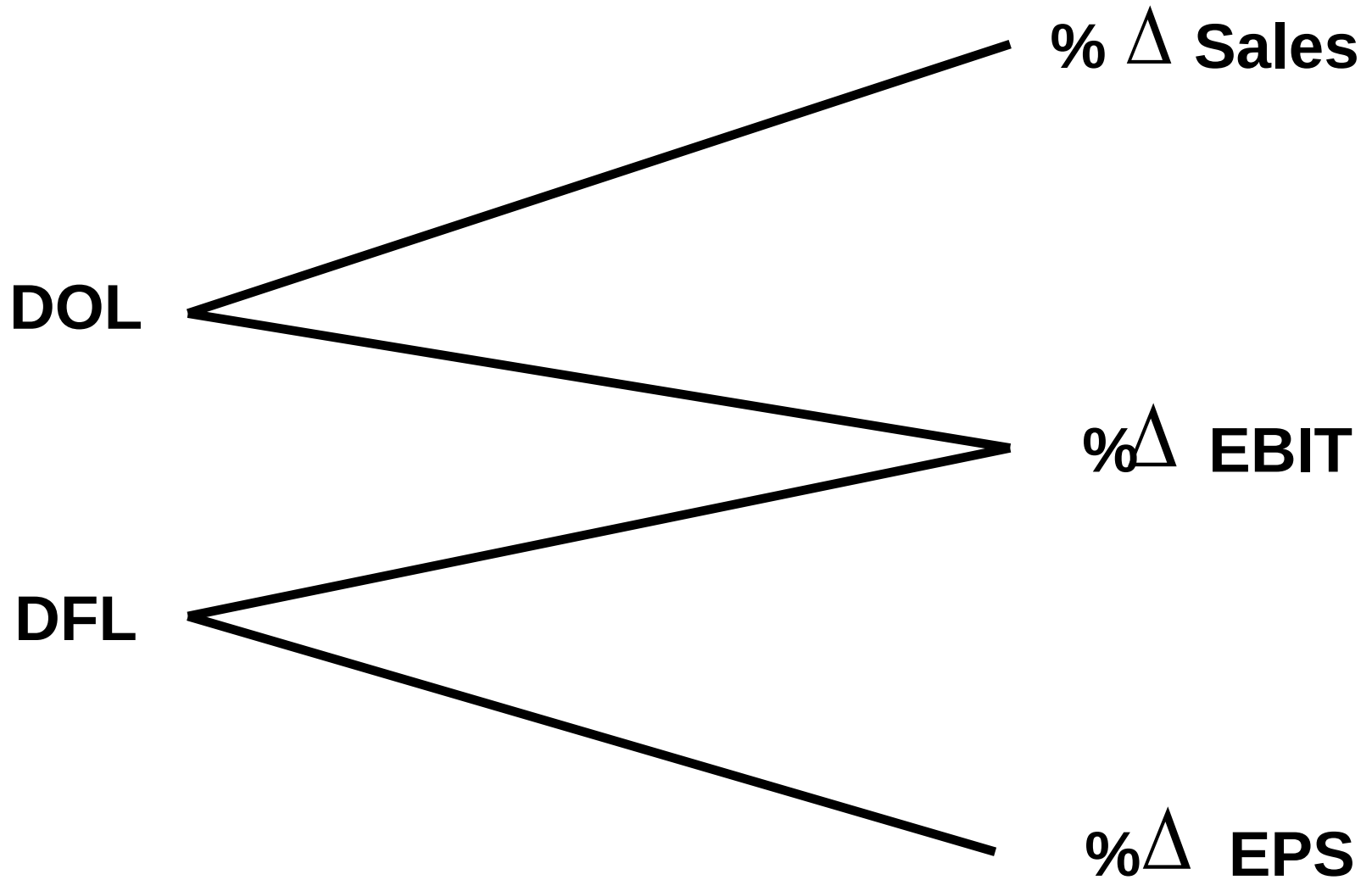
Total Leverage

- Total effect of operating and financial risk as measured through the financial and operating leverage measures can be stated as the combined / composite / total leverage.
- Computation of total leverage is a multiplication of financial leverage by the operating leverage. In other words this is the relationship between Contribution and profit before tax.

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{PBT}}$$

$$\text{Degree of Combined Leverage} = \frac{\Delta \text{EPS}}{\Delta \text{Sales}}$$

Total Leverage



Example

For the above illustrations, the combined leverage is:

Operating leverage x Financial leverage

1st Year: $1.25 \times 2 = 2.5$

2nd Year: $1.5 \times 1.8 = 2.7$

Degree of combined leverage is: $1.2 * 1.18 = 1.416$.

Inference

Risky situation: High operating leverage combined with high financial leverage.

Normal situation: High operating leverage but compensated by a low financial leverage or vice versa. However **when operating leverage is low and financial leverage is high, the EPS will be maximum.**

Ideal situation: Low operating leverage combined with low financial leverage.

Implications of Combined Leverage

- While interpreting the combined risk careful scrutiny is advocated since very high as well as very low risk positions are not suitable for business organizations.
- In the above illustration, the combined leverage or risk position for the firm has increased over time due to the high operating risk. The firm can be considered to be in the high risk category.

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Basic EPS - Continuing Operations	86.19	86.71	103.62	115.19	125.88
Basic EPS - Continuing Operations Growth	3.78%	0.61%	19.50%	11.16%	9.29%
Combined Leverage	1.64	1.62	1.61	1.69	1.57
Degree of Combined Leverage	0.53	0.13	1.16	0.63	1.36

EBIT – EPS Analysis

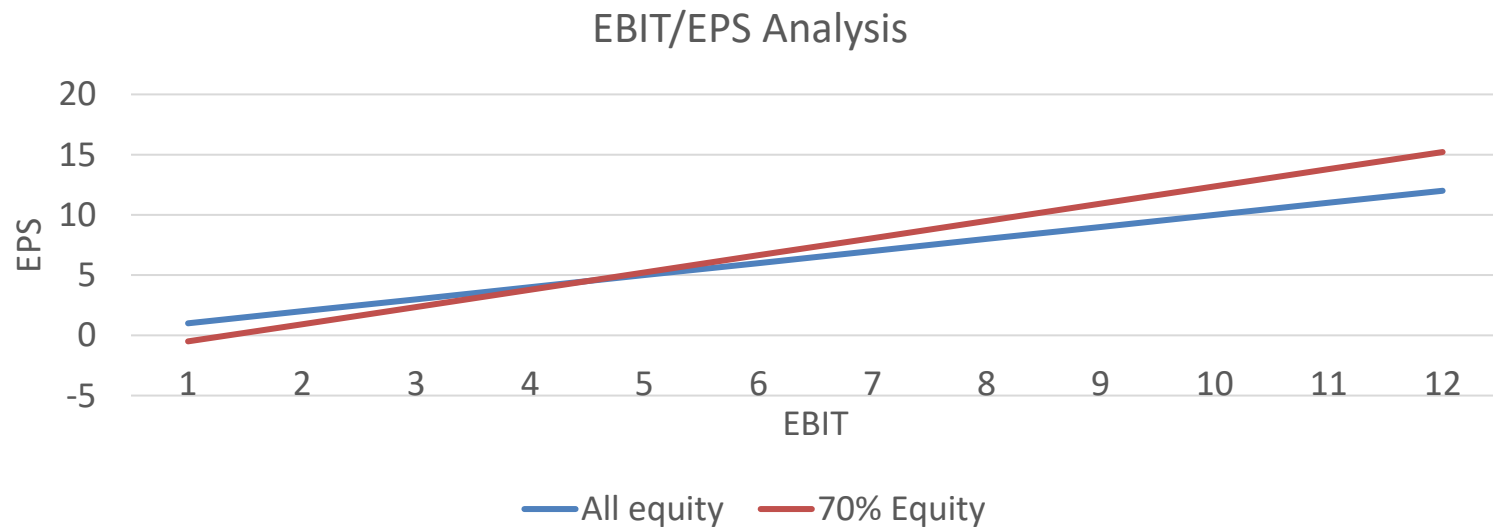
- $EPS = \frac{[(EBIT - \text{Interest}) \times (1-T)] - \text{Preference dividend}}{\text{Number of Shares}}$

Number of Shares

- Effect of financing alternatives on earnings per share
- Breakeven point is the level of debt at which the EPS is same for alternatives

EBIT/EPS Breakeven Point

- All Equity Number of Shares: 1000
- 10% Debt: 13500; Equity Number of Shares: 700
- EBIT variation 1000 to 12000



$$\text{BEP} = x/1000 = (x-13500)/700; x=4500$$